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Studierichting: Informatica
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Bereken in $\mathbb{C}$ door gebruik van goniometrische vormen: $(\sqrt{6} + i\sqrt{2})^{12}$

$$r = \sqrt{8}$$

$$\tan \theta = \frac{\sqrt{2}}{\sqrt{6}}$$

$$\theta = \text{Bgtan} \frac{\sqrt{3}}{3} = \frac{\pi}{6}$$

$$\left(\sqrt{8} \cdot \text{cis} \left(\frac{\pi}{6} \right) \right)^{12}$$

$$= \sqrt{8}^{12} \text{cis} \left(12 \cdot \frac{\pi}{6} \right)$$

$$= 8^6 \cdot \text{cis}(2\pi)$$

$$= 8^6 \cdot \text{cis}(0)$$

$$= 8^6 \cdot 1 + 8^6 i \cdot 0$$

$$= 262144$$

References